New bounds for open intersection arrays of completely regular codes in the Hamming graph H(n,3)

Sources. The open arrays and length ranges are taken from Chapter 7 of [1] and the online tables crcodes.xyz [2]; full references are given at the end.

Notation. {b; c} is the intersection array, $C_0 = C$, …, C_ρ are the cells of the distance partition (C_j is the set of vertices at distance j from the code), S is the quotient matrix (S_{jk} is the number of neighbours in C_k of a vertex of C_j). $[i_1,\dots,i_\rho]$ is the spectrum: the eigenvalues of S other than 2n are $\theta_{i_j} = 2n - 3i_j$. L–U is the length range according to Chapter 7 and crcodes.xyz. All computational steps of the proofs are carried out in exact arithmetic and come with checkable certificates.

The matrices $P_2(S)$, $P_3(S)$. Let A_w be the adjacency matrix of the "distance exactly w" graph of H(n,3). It is a polynomial of degree w in $A = A_1$:

$$A_2 = P_2(A) = (A^2 - A - 2n\cdot I)/2, \quad A_3 = P_3(A) = (A\cdot A_2 - 2A_2 - 2(n-1)\cdot A)/3$$

(this is the recurrence $A\cdot A_w = (w+1)A_{w+1} + w(q-2)A_w + (n-w+1)(q-1)A_{w-1}$ with $q = 3$). Hence the colouring of the vertices by the cells C_0, …, C_ρ is also a perfect colouring for A_w, with quotient matrix $P_w(S)$: **$(P_w(S))_{jk}$ is the number of vertices of C_k at distance exactly w from a fixed vertex of C_j**. These are the matrices S_w of Chapter 1; $P_2(S)$ and $P_3(S)$ are the cases w = 2, 3.

1. Methods

(F) Orbits of a vertex pair and the dual Farkas problem

Take two vertices $x = 0^n$, $y = 1^d0^{n-d}$ of known colours (usually both are codewords) and partition H(n,3) into the orbits of the stabiliser of the pair: the orbit of a vertex z is (α, β, γ, δ), where α, β, γ are the numbers of ones, zeros and twos among the first d coordinates of z, and δ is the number of nonzero coordinates among the rest. The partition into orbits is perfect (quotient matrix R). The fractions H_{Oc} of vertices of colour c in the orbit O satisfy RH = HS, H ≥ 0, every row sums to 1; the colours of the singleton orbits {x} and {y} are fixed. Infeasibility of this system is proved via Farkas' lemma by an integer dual table Y (the certificate). It is found by exact rational LP, and checking the certificate is a finite computation.

(FS) Subcube orbits: {13,2;4,8} at n = 7

The scheme is the same, but instead of a vertex pair we fix a codeword x and three coordinates l_1, l_2, l_3. The **subcube** is the 3-face through x: the 27 vertices $x + a\cdot e_{l_1} + b\cdot e_{l_2} + c\cdot e_{l_3}$, $(a, b, c) \in Z_3^3$.

- **Classes.** The class of a vertex z is the pair (q, e): $q = (z_{l_1} - x_{l_1}, z_{l_2} - x_{l_2}, z_{l_3} - x_{l_3}) \in Z_3^3$ is the position of the "projection" of z in the subcube, and $e \in \{0, \dots, 4\}$ is the number of the remaining $n - 3 = 4$ coordinates where $z \neq x$. These are the orbits of the subgroup of Aut H(7,3) fixing x and each of the coordinates l_1, l_2, l_3: permutations of the other four coordinates and, in each of them, swapping the two values different from x_k. There are $27 \cdot 5 = 135$ classes, $|(q, e)| = C(4, e)\cdot 2^e$.
- **Main difference from pair orbits:** for e = 0 a class consists of a single vertex of the subcube, so the colours of all 27 vertices of the subcube can be fixed individually, not just two.
- **Quotient matrix.** A vertex of class (q, e) has one neighbour in each of the classes $(q \pm e_j, e)$ (j = 1, 2, 3: changing coordinate l_j), e neighbours in (q, e − 1), e neighbours in (q, e) and 2(4 − e) neighbours in (q, e + 1). Then the same system RH = HS with fixed rows and the same kind of certificate.
- **Case tree.** $a_0 = 1$, so x has exactly one neighbour in the code; choose l_1 so that it is $x + e_{l_1}$ or $x + 2e_{l_1}$ (two branches). Then the colours of the subcube vertices (0,0,1), (0,0,2), (0,1,0), … are enumerated one by one; a branch is cut as soon as the system with the colours fixed so far is infeasible. This gives a tree with 146 leaves, each closed by its own certificate.

(E) Existence

A code is the preimage of a perfect colouring $D \subset F_3^m$ of the Cayley multigraph $Cay(F_3^m, \pm\text{columns of H})$, where H is an m × n parity-check matrix (Lemma 1.9, Prop. 1.36). If D is perfect for two sets of columns, then it is perfect for their union, and the matrices add up (concatenation lemma).

(M) The last eigenvalue $\theta = \theta_{i_\rho}$

Let $Su = \theta u$ and $f(x) = u_{d(x,C)}$; then $Af = \theta f$. **Vanishing on faces:** if $\theta = \theta_i$, then the sum of f over any face of dimension > n − i is zero. A face of dimension d consists of the 3^d vertices with d free coordinates and the others fixed. For $i = i_\rho$ and length $n = i_\rho + d - 1$, all d-faces have zero sum.

The labelling problem. In the tests Sq, Cu, P9, F4 we take a vertex x of cell j and a face through x. Its vertices are labelled by their distances to the code $g(a) = d(x + \dots, C)$. The labelling must satisfy: g(x) = j; the labels of two adjacent vertices differ by at most 1; the sums of u_g over the relevant faces vanish. In addition, some labels can be fixed:

- **"path"**: if all $b_i > 0$, there is a path $x \to z \to y$ through the cells j, j+1, j+2. The vertex y differs from x in exactly two coordinates (otherwise it would be a neighbour of x, and labels of neighbours differ by at most 1), so y lies in a 2-face through x; similarly a path of length 3 lies in a cube;
- **"lines"**: if $2n - S_{jt} + K \le n$, then at least K lines through x lie entirely (apart from x) in cell t; their directions are taken as the first axes of the face.

If no admissible labelling exists, there is no code. The enumeration of labellings was carried out by computer.

- **(L) Lines, $n = i_\rho$, $\theta = -n$.** Ratio condition: for each $i < \rho$ either $2u_i + u_{i+1} = 0$ or $u_i + 2u_{i+1} = 0$. It rules out all 52 rows of the $q = 3$ tables for which $n = i_\rho$ lies in the range L–U (the remaining such rows are known codes with $L = U = n$, for which the condition holds).
- **(Sq) Squares, $n = i_\rho + 1$.** The face is the 3×3 grid $\{x + a\cdot e_l + b\cdot e_m\}$, and the sum of u_g over it is 0. Example: {10,5;4,8} at $n = 7$, $u = (25, -20, 16)$, $x \in C$. The path $x \to z \to y$ through cells 0, 1, 2 gives a square with labels 0 and 2. If it contains n_c vertices of cell c, then $n_0 + n_1 + n_2 = 9$ and $25n_0 - 20n_1 + 16n_2 = 0$, i.e. $5n_0 + 4n_2 = 20$. The solutions (4, 0) and (0, 5) lose one of the labels 0, 2 — a contradiction.
- **(Cu) Cubes, $n = i_\rho + 2$.** The face is the 27 vertices $\{x + a\cdot\alpha e_{l_1} + b\cdot\beta e_{l_2} + c\cdot\gamma e_{l_3} : a, b, c \in Z_3\}$, and the sum over the whole cube vanishes. Fixings: "path" (a path of length 2 or 3) or "lines"; in addition we use the neighbour bound: inside the cube a vertex of cell c has at most $S_{cc'}$ neighbours of cell c'.
- **(P9) Nine planes of a cube, $n = i_\rho + 1$.** Here $\theta = \theta_{n-1}$, so every 2-face of the graph has zero sum, not only the squares through x. Take the cube of 27 vertices $x + a\cdot e_{l_1} + b\cdot e_{l_2} + c\cdot e_{l_3}$ (a, b, c $\in Z_3$), where x is a codeword. It has 9 planes parallel to its faces: {a = 0}, {a = 1}, {a = 2}, {b = 0}, {b = 1}, {b = 2}, {c = 0}, {c = 1}, {c = 2}. Each of them is a 2-face of H(n,3) (two coordinates free, the rest fixed). Only three pass through x: {a = 0}, {b = 0}, {c = 0}. The other six are parallel to them and pass through neighbours of x. The test requires all 9 sums of u_g to vanish simultaneously, with g(0,0,0) = 0, the condition "neighbours differ by at most 1" and the neighbour bound. The square test (Sq) checks the squares through x separately, so P9 is stronger. It rules out {32,18;10,21} at $n = 19$ ($u = (16, -11, 7)$) and {44,42,10;1,12,35} at $n = 22$ ($u = (44, -19, 8, -10)$).
- **(F4) 4-faces, $n = i_\rho + 3$.** 81 vertices, zero sum over the whole face, a path $x \to \ldots$ through cells $0 \to 1 \to 2 \to 3 \to 4$. Non-existence of a labelling is proved by a SAT solver, and its LRAT certificate was checked independently. For arrays of the form $b_j = 5(\rho - j)$, $c_j = 4j$ there is also a hand proof.
- **(CC) Cube counting, $n = i_\rho + 2$.** Here we consider all cubes and squares through a vertex x of cell j at once, not a single cube. Three facts are used:

  1. every vertex at distance 3 from x lies in exactly one 3-face through x (the free coordinates are the three where it differs from x);
  2. every vertex at distance 2 lies in exactly one 2-face through x, and every 2-face through x lies in exactly $n - 2$ cubes through x;
  3. the number of vertices of cell k at distance 2 from x is $(P_2(S))_{jk}$, and at distance 3 it is $(P_3(S))_{jk}$.

  All admissible labellings of the $3 \times 3 \times 3$ cube (zero sum, g(x) = j, neighbours differ by at most 1) are grouped into "patterns": the types of the three lines through x, the contents of the interiors of the three squares through x, and the contents of the 8 vertices at distance 3. For every possible assignment of types to the n lines through x the unknowns are the numbers of cubes and squares of each pattern (nonnegative). Facts 1–3 give linear equations: the sum over cubes of the contents at distance 3 equals row j of $P_3(S)$; likewise for squares and $P_2(S)$; every square is counted in $n - 2$ cubes. If the system is infeasible over $\mathbb{Q}$ for all assignments of line types, there is no code. The systems are solved by exact rational LP, and for each infeasible system a Farkas vector is found. For {29,18;10,24} and {30,7;18,26} at $n = 20$ a single cube suffices (two lines in cell t and a third with a vertex of cell $t \pm 2$). In the remaining 17 cases the certificate consists of Farkas vectors for all assignments of line types; the full enumeration of cube labellings and the checking of the certificates were carried out by computer in exact arithmetic.

2. Updated bounds

In the "method" column, the length n excluded by each test is given in parentheses; "L (n)" means the length $n = i_\rho$ is excluded by lines (L). The L–U column is the range according to crcodes.xyz [2] (Chapter 7 [1]). The table has two parts: in the first the lower bound is raised (a bold number in the "now" column means the minimum length is determined, L = U); the second contains the array {25;20}, whose upper bound is lowered to 17 by a construction (E). All other upper bounds are unchanged.

array	L–U (crcodes.xyz)	now	method (excluded length)
Lower bound raised			
{14,7;2,4}	7–8	**8**	F (7); Sq
{21,14,7;2,4,6}	11–12	**12**	F (11); Cu
{28,21,14,7;2,4,6,8}	14–16	**16**	F (14); Cu (14); F4 (15)
{10,5;4,8}	6–8	**8**	L (6), Sq (7)
{15,10,5;4,8,12}	9–12	**12**	L (9), Sq (10), Cu (11)
{20,15,10,5;4,8,12,16}	12–16	**16**	L (12), Sq (13), Cu (14), F4 (15)
{30,15;12,24}	18–20	**20**	L (18), Sq (19)
{45,30,15;12,24,36}	27–30	**30**	L (27), Sq (28), Cu (29)
{20,10;8,16}	12w–16	14–16	L (12), Sq (13)
{30,20,10;8,16,24}	18–24	21–24	L (18), Sq (19), Cu (20)
{40,30,20,10;8,...,32}	24–36	28–36	L (24), Sq (25), Cu (26), F4 (27)
{25,20,15,10,5;4,...,20}	15–20	19–20	L (15), Sq (16), Cu (17), F4 (18)
{35,28,21,14,7;2,...,10}	18–20	19–20	F4 (18)

{50,40,30,20,10;8,...,40}	30–40	34–40	L (30), Sq (31), Cu (32), F4 (33)
{40,20;16,32}	24–32	26–32	L (24), Sq (25)
{42,28,14;13,26,39}	27–33	31–33	L (27), Sq (28), Cu (29), F4 (30)
{48,32,16;11,22,33}	27–33	31–33	L (27), Sq (28), Cu (29), F4 (30)
{28,14;13,26}, {32,16;11,22}, {34,17;10,20}	18–22	21–22	L (18), Sq (19), Cu (20)
{38,19;8,16}	19–22	21–22	F (19); Cu (20)
{40,20;7,14}	20–22	21–22	Cu (20)
{24,20;4,6}	13w–?	≥ 15	F (13); Cu (14)
{36,30;6,9}	19w–?	≥ 21	F (19); Cu (20)
{48,40;8,12}	25w–?	≥ 27	F (25); Cu (26)
{20,10;2,13}	10–?	≥ 11	F (10)
{38,10;11,22}	19–?	≥ 20	F (19)
{13,2;4,8}	7–?	≥ 8	FS: subcube orbits (7)
{30,22;8,21}, {34,33,8;2,12,28}	18–?	≥ 21	L (18), Sq (19), Cu (20)
{44,40,30,4,2;2,4,24,40,44}	22–?	≥ 25	L (22), Sq (23), CC (24)
{28,5;4,26}	14–?	≥ 15	CC (14)
{28,9;2,24}	14–?	≥ 15	CC (14)
{44,42,10;1,12,35}	22–?	≥ 23	P9 (22)
{37,14;10,20}	19–?	≥ 20	Sq (19)
{32,18;10,21}	18–?	≥ 20	L (18), P9 (19)
{20,18;4,12}	12–?	≥ 15	L (12), Sq (13), CC (14)
{22,20;4,8}	12–?	≥ 15	L (12), Sq (13), CC (14)
{29,10;8,25}	15–?	≥ 17	L (15), Sq (16)
{29,18;10,24}	18–?	≥ 21	L (18), Sq (19), CC (20)
{30,7;18,26}	18–?	≥ 21	L (18), Sq (19), CC (20)
{30,20;9,22}	18–?	≥ 20	L (18), Sq (19)
{30,27;6,18}	18–?	≥ 21	L (18), Sq (19), CC (20)
{33,4;18,26}	18–?	≥ 21	L (18), Sq (19), CC (20)
{33,10;14,24}	18–?	≥ 21	L (18), Sq (19), CC (20)
{33,30;6,12}	18–?	≥ 21	L (18), Sq (19), CC (20)
{34,9;14,24}	18–?	≥ 20	L (18), Sq (19)
{34,26;7,14}	18–?	≥ 20	L (18), Sq (19)
{35,22;8,16}	18–?	≥ 20	L (18), Sq (19)
{38,28;5,28}	21–?	≥ 24	L (21), Sq (22), CC (23)
{40,36;8,24}	24–?	≥ 27	L (24), Sq (25), CC (26)
{42,20;9,28}	21–?	≥ 23	L (21), Sq (22)
{42,26;7,24}	21–?	≥ 24	L (21), Sq (22), CC (23)
{42,26;13,42}	27–?	≥ 29	L (27), Sq (28)
{44,40;8,16}	24–?	≥ 27	L (24), Sq (25), CC (26)
{48,20;9,40}	24–?	≥ 26	L (24), Sq (25)
{48,22;11,48}	27–?	≥ 29	L (27), Sq (28)
{48,35;4,30}	24–?	≥ 27	L (24), Sq (25), CC (26)
{30,20,12;3,4,30}	15–?	≥ 18	L (15), Sq (16), CC (17)
{36,22,2;2,27,28}	18–?	≥ 21	L (18), Sq (19), CC (20)
{36,26,3;4,18,30}	18–?	≥ 21	L (18), Sq (19), CC (20)
{48,30,7;9,20,30}	24–?	≥ 26	L (24), Sq (25)
Range narrowed from above: new code (E)			
{25;20}	16–19	16–**17**	E: upper bound lowered to 17 (§3)

For rows of the form "L (i_ρ), Sq ($i_\rho + 1$), CC ($i_\rho + 2$)" the lengths i_ρ, $i_\rho + 1$ and $i_\rho + 2$ are excluded, i.e. L ≥ $i_\rho + 3$. The CC test (cube counting) rules out 17 lengths, including those for {28,5;4,26} and {28,9;2,24} (L ≥ 15, while crcodes.xyz has L = 14).

3. Dual Farkas certificates and existence

Certificates (F). In all cases x and y are codewords (colour 0). The values Σ_O m(O) > 0 were recomputed in exact arithmetic.

array	n	d(x,y)	third word	Σ m(O)
{14,7;2,4}	7	2	—	2
{21,14,7;2,4,6}	11	11	—	30
{28,21,14,7;2,4,6,8}	14	2	—	321
{24,20;4,6}	13	1	—	42
{36,30;6,9}	19	1	—	1131

{48,40;8,12}	25	1	—	436
{20,10;2,13}	10	2	colour 0 / 1 / 2	825 / 758 / 14
{38,10;11,22}	19	2	colour 0 / 1 / 2	4846 / 1290 / 4596
{38,19;8,16}	19	2	colour 0 / 1 / 2	38 / 4572 / 790
{13,2;4,8}	7	—	subcube orbits, m = 3	tree of 146 cases

The third word (0,0,d,0) is a vertex that differs from both x and y in all d positions. All three of its colours are enumerated, with a separate certificate for each.

The last row, {13,2;4,8}, is obtained by method (FS): a tree of 146 cases, each with its own certificate.

Existence (E): {25; 20} in H(17,3). crcodes.xyz gives 16–19; the upper bound is lowered to **17**. Take $D \subset F_3^4$, $|D| = 36$, and two sets of columns: B (7 columns) and O (the 10 points of the ovoid $x_0^2 + x_1^2 + x_2^2 = x_3^2$ in PG(3,3)). The colouring $(D, F_3^4 \setminus D)$ is perfect for B (array {10; 8}, n = 7) and for O (the Brouwer–Haemers graph, array {15; 12}, n = 10). Hence the parity-check matrix H = [B | O] gives the array {25; 20} at n = 17, $|C| = 36 \cdot 3^{13}$.

References

[1] J. H. Koolen, D. S. Krotov, W. J. Martin. Completely Regular Codes: Tables of Small Parameters for Binary and Ternary Hamming Graphs. Chapter 7 in "Completely Regular Codes in Distance Regular Graphs". CRC Press, 2025. https://doi.org/10.1201/9781003393931.

[2] D. S. Krotov, W. J. Martin, J. H. Koolen. Completely regular codes: tables of small parameters (online tables). https://crcodes.xyz.